4. Genetic Algorithms

In optimization, there are 3 must needed requirements for all problems.

1. Decision Variables - What we need to optimize
2. Objective function - What we expect at last
3. Constraints - Limitations on what we can do or use

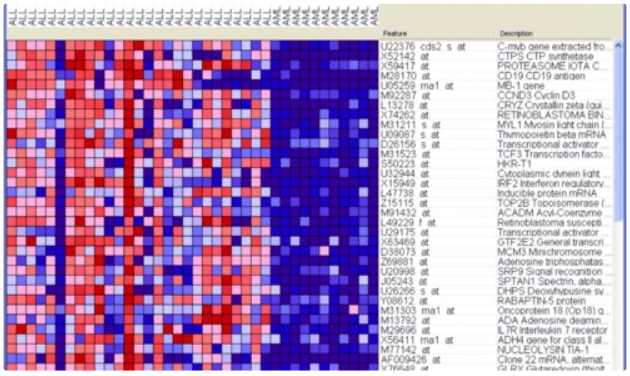
Usually, when we know the first derivative($f'(x)$) of a function, we can find optimum value by finding the values of x which makes $f'(x)$ = 0.

But in most of the cases , we can't find the $f'(x)$. So we have to use other ways to solve those questions.

For that, scientists found that we can solve these questions using how nature solves problems and how evolution made species survived throughout tough times. One of them is Genetic Algorithms.

What is a genetic algorithm?

First, what is a gene?



A gene is a collection of nucleotides. They are organized in a pattern and that pattern decides whether this person is suitable to survive or now.

In the same way, when people create genetic algorithms, they select a feasible solution for the problem and encode them in a finite characters.

In GAs, objective function is called "*fitness Function*".

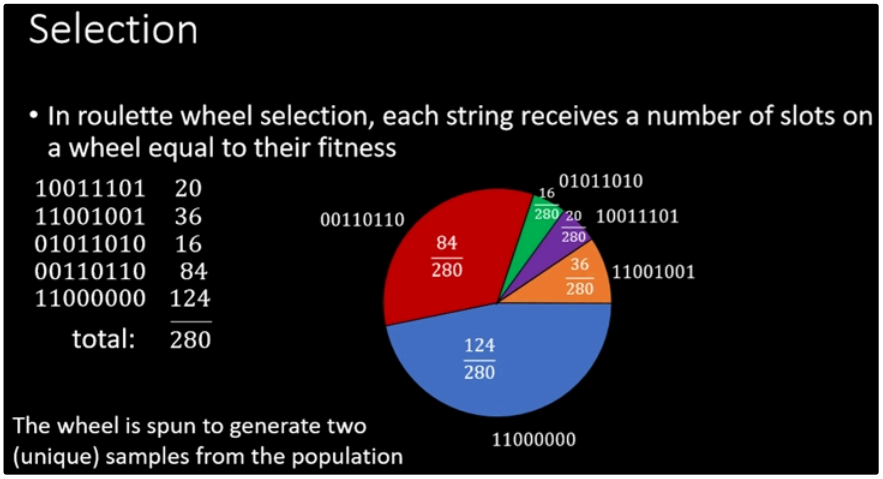
Each string inside a feasible solution is called as a *phenotype*. So each phenotype has a fitness value.

Genetic Operators

When we use genetic Algorithms, we use 3 operators to solve the problems. This can be easily explained using how humans receive genetic information from their parents.

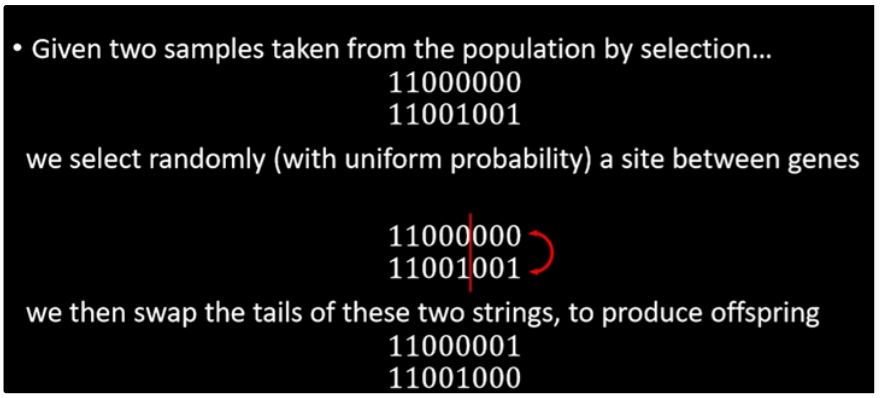
Selection

First, we have to select 2 people to breed. In the same way, we choose 2 answers from a huge number of answers. It's done through a probability proportionate to fitness of those. For that, we usually use roulette wheel selection and fitness value of each gene.



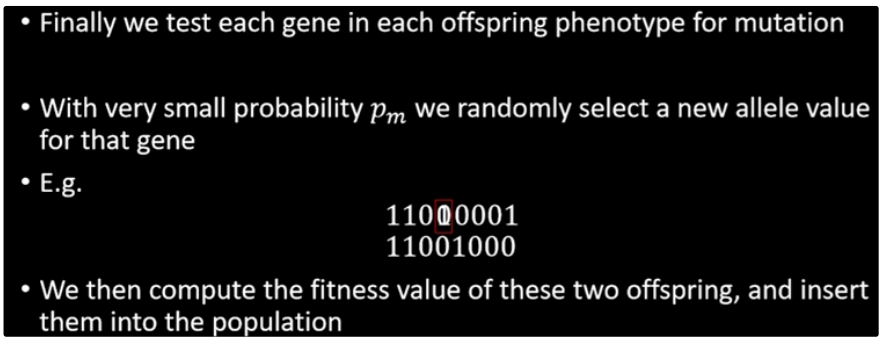
Crossover

After we chose 2 samples, we need to make the result a product of both parents (Like we have features from both mother and father). Usually what we do is find a separation randomly and make 2 new child answers from them.



Mutation

For a difference from both parents, we change the value of each child (offspring) answer, maybe a higher value or a smaller value.



After all these, we destroy answers with least fitness and do this same operations again and again until some conditions are met (Termination condition).

- Common terminating conditions are:
  - A solution is found that satisfies minimum criteria
  - Fixed number of generations reached
  - Allocated budget (computation time/money) reached
  - The highest ranking solution's fitness is reaching or has reached a plateau such that successive iterations no longer produce better results
  - Manual inspection
  - Any Combinations of the above